

AJAI KUMAR N

SOFTWARE ENGINEER · BACKEND DEVELOPER

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SUMMARY

Computer Science & Engineering student graduating May 2027 with hands-on experience building backend systems and REST APIs for business clients. Built ERP and fleet management platforms using Java, Spring Boot, Python, FastAPI, and PostgreSQL.

EDUCATION

B.Tech Computer Science & Engineering (AI)

Karunya Institute of Technology and Sciences

Expected: May 2027 | Started: June 2023

TECHNICAL SKILLS

Languages: Java, Python

Backend: Spring Boot, FastAPI, SQLAlchemy, REST APIs

Web: React.js, TypeScript

Mobile: Flutter

Databases: PostgreSQL, MySQL, MongoDB

Caching & Real-Time: Redis, WebSockets

Tools: Git / GitHub, Postman

WORK EXPERIENCE

Spring Boot Backend Intern · Icanio Tech School

May – Jun 2025

- Built backend modules for a College ERP system using Java and Spring Boot
- Developed RESTful APIs for academic management backed by a PostgreSQL database
- Implemented database modules and CRUD operations for academic records

Web Development Intern · LifeChangersInd

Jun – Jul 2024

- Built attendance tracking and academic record management modules for a School ERP Management System
- Implemented role-based access control for three user roles - administrators, teachers, and students

SELECTED ENGINEERING PROJECTS

Independent Software Developer

Jan 2024 – Present

Yamini Infotech ERP

Dec 2025 – Jan 2026

Python · FastAPI · React · Flutter · PostgreSQL · MongoDB

- Designed and built an ERP system with a Python/FastAPI backend and PostgreSQL database, serving workforce management, service operations, and CRM for a business client
- Implemented GPS live tracking, attendance management, CRM, and service operations modules; used MongoDB for GPS tracking data

Kavya Transports ERP

Jan 2026 – May 2026

Python · FastAPI · SQLAlchemy · React · TypeScript · Flutter · PostgreSQL · MongoDB · Redis · WebSockets

- Built a fleet management platform using Python/FastAPI and SQLAlchemy, covering vehicle tracking, trip monitoring, billing, and transport operations
- Designed REST API endpoints and relational database architecture in PostgreSQL for fleet operations; integrated MongoDB for GPS and log data, and WebSockets for real-time updates

UZHAVAN AI — Agricultural Decision-Support System

- Built a Tamil-first agricultural decision-support system combining TNAU-grounded knowledge, weather-aware recommendations, crop and disease intelligence, and agricultural scheme information retrieval
- Developed and evaluated ML pipelines for crop classification and yield prediction, achieving F1 = 0.839 for crop classification and RMSE = 749 kg/ha for yield prediction